



● AINS HACKATHON 2026 · AINS 4.0 × Vectors

SENTINEL

The reliability layer for enterprise AI agents

Use Cases 1 + 2 + 3, unified — continuous evaluation, deterministic replay, and a real Atlassian Rovo agent we instrument and dogfood.

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Enterprises ship agents they cannot trust

The problem

- **Non-deterministic.** The same instruction yields different tool calls and outputs each run.
- **Failures are silent.** It finishes its calls — yet mis-triages a ticket or overwrites a page.
- **Exact-match tests fail.** They assume one correct output; agent trajectories have many.
- **Failures don't reproduce.** Re-running changes the output and re-fires side effects.

12/80

tickets mis-
routed, *zero*
alerts

0

reproducible
replays from a
log

?

which step
actually failed

3 wk

later: "what
did it see?"

No continuous evaluation · no deterministic debugging · no audit trail for production AI agents.

Who we build for

Target users

AI platform owners

Need *continuous* evaluation and drift detection on every production run — not a one-shot pre-launch check.

SRE & devtools teams

Run agents in prod and must debug failures deterministically, without re-firing real side effects.

Marketplace AI vendors

Ship on Atlassian Forge / Rovo and need auditable verdicts to prove an agent is safe to release.

Risk & compliance

Need a tamper-evident record of exactly what an agent saw and did, weeks after the fact.

Scoped to **incident-response** agents on **Atlassian JSM** — a specific agent type and task domain.

One platform, three jobs

Proposed solution

UC2 · Flight Recorder

Transparently **captures** every LLM and tool call; replays a run **deterministically** with zero live calls, plus bisect and mid-replay injection.

UC1 · Eval Engine

Judges those traces into auditable verdicts: multi-level scores, per-step failure attribution, pass^k reliability, and drift.

UC3 · Atlassian Agent

A **real** Rovo incident-response agent on Forge — instrumented by UC1+UC2; its verdicts land back in Jira as Incidents.

“We built the reliability infrastructure a Marketplace AI vendor needs — and dogfooded it on a real Atlassian agent.”

AI is the mechanism, not a feature

Core AI mechanism

Remove the models and nothing works — every output is reasoning over live signals, not keyword search.

- **Semantic retrieval** — BGE-768 finds similar incidents and runbooks, with attribution.
- **Structured generation** — Llama-3.1-8B drafts the RCA and duplicate verdict as typed objects.
- **LLM-as-judge** — scores 4 rubric dimensions, run twice with order reversed to calibrate bias.
- **Safety classifier** — Llama-Guard-3 pre-filters every transcript.
- **Failure attribution** — a DAG blames retrieval, planning, or execution.
- **Evaluating the evaluator** — Cohen's κ vs. a gold set; centroid **drift** catches shifts a pass/fail can't.

bge-base-en-v1.5

llama-3.1-8b-instruct

llama-guard-3-8b

on Cloudflare Workers AI

Every verdict is traceable

Explainability & auditability

Per-step attribution

“Step 2, retrieval, returned no results, confidence 0.7” — not just “it failed”.

Confidence & self-eval

The judge scores its own verdict and **flags for a human** below threshold.

Tamper-evident audit

Hash-chained, HMAC-signed, write-ahead record of every step the agent took.

Deterministic replay

Re-run the exact trajectory from tape — inspect any prompt, inject a fix, see it diverge.

A non-AI engineer can read the verdict and act: *what it should have done, what it did, where it diverged, and the next step.*

We evaluate the evaluator

Evaluation & rigour

- **pass^k (k=8)**, the τ -bench metric — all 8 trials must pass; exposes inconsistency pass@1 hides.
- **Cohen's κ** — judge-vs-human agreement on a gold set, chance-corrected (a constant “pass” scores ≈ 0).
- **Drift** — pass-rate, per-dimension, and semantic output-embedding shift across run windows.
- **Non-determinism addressed** — calibration flips an order-sensitive verdict to *uncertain*, not a guess.

100%

pass@1
(single-trial)

0%

pass⁸: reliability gap

0.60

Cohen's κ
(moderate)

70%

consistency rate

Test set: 100 seeded incidents, 10 root-cause categories. Full numbers in the Evaluation Report.

Why Sentinel wins

Value proposition

Engineering depth

50%

Infrastructure-grade: transport-level interception, hash-chained audit, calibrated judging — the AI *is* the system.

Prototype quality

25%

End-to-end demo on realistic synthetic data, live and deployed, with a graceful mock fallback.

Explainability

15%

Attribution, confidence, audit trail, and deterministic replay on every verdict.

Eval & rigour

10%

A real protocol: pass^k , Cohen's κ , drift — and we evaluate the evaluator.

Bonus: an open **OTel GenAI replay** spec and an **MCP audit-trail** proposal — documented protocol contributions.

● LIVE & DEPLOYED

Reliability infrastructure, dogfoodéd on a real agent.

Three Python services on Azure behind Cloudflare Tunnel · a Forge Rovo agent installed on Jira & Confluence · a Next.js mission-control dashboard.

github.com/ahmedxsaad/AINS · dashboard.ahmedxsaad.me